Vision Relation Transformer for Unbiased Scene Graph Generation

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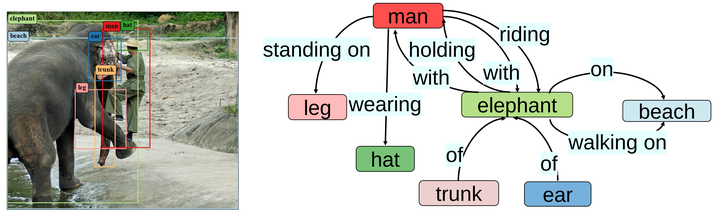
1. Introduction

Scene Graphs

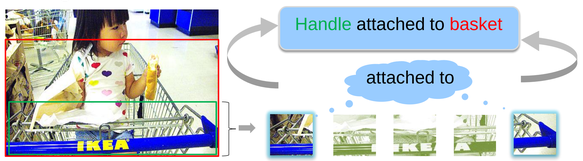


Highlights

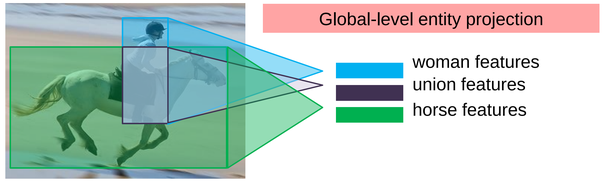
- We propose the **Vision rElation TransfOrmer** (**VETO**) to address **two challenges** in SGG.
- We propose a **Mutually Exclusive ExperT** (**MEET**) learning strategy to address **another challenge** in SGG.
- We extensively evaluate our model on two benchmark SGG datasets Visual Genome and GQA.
- VETO and VETO + MEET outperform the state of the art while being **10 times smaller.**

Challenges in Scene Graph Generation

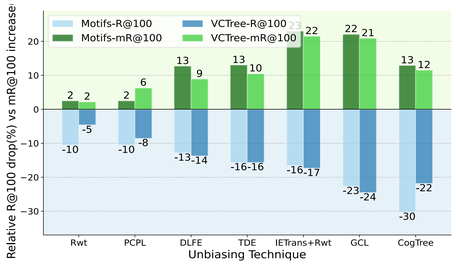
- Challenge 1: Local-level information loss



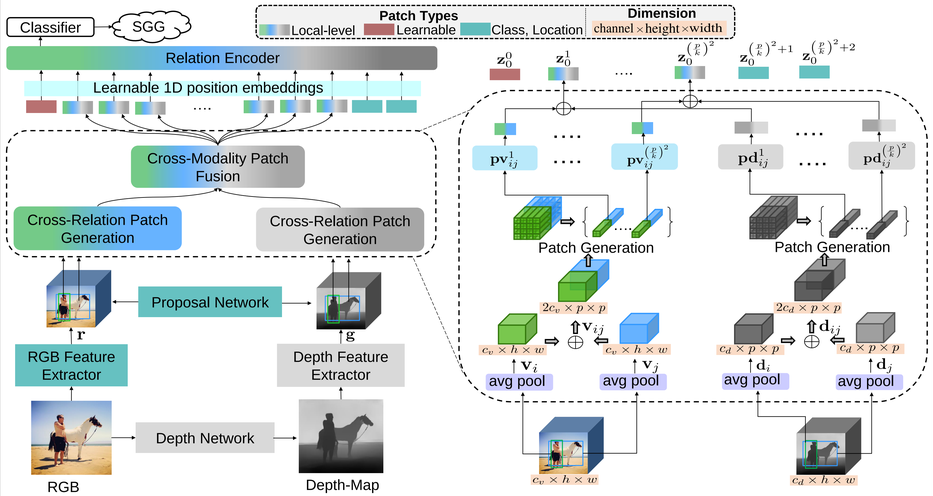
- Challenge 2: Parameter heavy models



- Challenge 3: Debiased models are still biased



2. Vision Relation Transformer (VETO)



- An object detector yields entity proposals and entity features resulting in visual cues.
- A depth-map is generated from RGB input from which geometric cues are extracted.
- Cross-Relation and Cross-Modality Patch Generation fuse the modalities and generate *multiple local-level patches* for each entity instead of a single global-level patch per entity, addressing **challenges 1 and 2**.

3. Mutually Exclusive Experts (MEET)

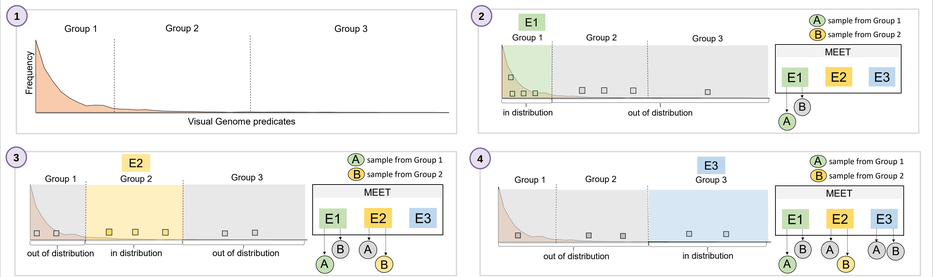


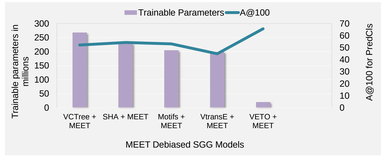
Illustration of MEET for a 3 expert setup

- Split the dataset sorted based on its predicate frequency distribution into multiple groups.
- Each expert is trained on in-distribution samples from a specific group and evenly sampled out-of-distribution samples from other groups.
- Out-of-distribution predictions highlighted in grey circles are discarded.
- MEET learning addresses **challenge 3**.

4. Experiments

Model	PredCls			SGCls		
	R@k: 50/100	mR@k: 50/100	A@k: 50/100	R@k: 50/100	mR@k: 50/100	A@k: 50/100
VTransE	65.7 / 67.6	14.7 / 15.8	40.2 / 41.7	38.6 / 39.4	8.2 / 8.7	23.4 / 24.1
VTransE + **MEET** (ours)	58.3 / 64.9	18.3 / 24.9	38.3 / 44.9	35.8 / 39.1	12.8 / 16.7	24.3 / 27.9
Motifs + EMB	65.2 / 67.3	18.0 / 19.5	41.6 / 43.4	39.2 / 40.0	10.2 / 11.0	24.7 / 25.5
Motifs + IETrans + Rwt	48.6 / 50.5	35.8 / 39.1	42.2 / 44.8	29.4 / 30.2	21.5 / 22.8	25.5 / 26.5
Motifs + **MEET** (ours)	67.4 / 72.7	25.3 / 33.5	46.4 / 53.1	40.5 / 43.2	19.0 / 23.7	29.8 / 33.5
VCTree	65.4 / 67.2	16.7 / 18.2	41.1 / 42.7	**46.7 / 47.6**	11.8 / 12.5	29.3 / 30.1
VCTree + EMB	64.0 / 65.8	18.2 / 19.7	41.1 / 42.8	44.7 / 45.8	12.5 / 13.5	28.6 / 30.0
VCTree + IETrans + Rwt	48.0 / 49.9	37.0 / 39.7	42.5 / 43.5	30.0 / 30.9	19.9 / 21.8	25.0 / 26.4
VCTree + **MEET** (ours)	62.0 / 69.8	25.5 / 34.5	43.8 / 52.2	35.4 / 39.2	14.5 / 18.6	25.0 / 28.9
SHA	64.3 / 66.4	18.8 / 20.5	41.5 / 43.5	38.0 / 39.0	10.9 / 11.6	24.5 / 25.3
SHA + GCL	35.1 / 37.2	41.6 / 44.1	38.4 / 40.7	22.8 / 23.9	**23.0 / 24.3**	22.9 / 24.1
SHA + **MEET** (ours)	66.3 / 72.4	28.0 / 36.2	47.2 / 54.3	37.9 / 41.2	16.1 / 20.7	27.0 / 31.0
VETO (ours) + Rwt	61.9 / 63.9	33.1 / 35.1	47.5 / 49.5	35.1 / 36.3	16.3 / 17.4	25.7 / 26.9
VETO + MEET (ours)	**74.0 / 78.9**	**42.0 / 52.4**	**58.0 / 65.7**	41.1 / 44.0	22.3 / **27.4**	**31.7 / 35.7**

Table 1. Recall (R), mean Recall (mR), and their average (A) on Visual Genome (the higher, the better). The best results before MEET debiasing are underlined and after MEET debiasing are highlighted in **bold.**



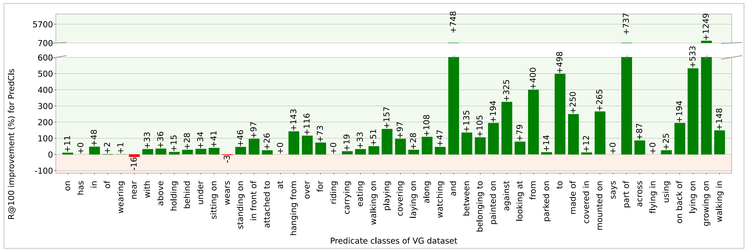
Plot 1. No. of trainable parameters (Mio.) vs. performance (A@100) of MEET debiased SGG models. VETO outperforms prior work and is ~ 10× lighter.

VETO Components				SGCls		
L	CR	CM	D	R@k: 50 / 100	mR@k: 50 / 100	A@k: 50 / 100
✗	✗	✗	✗	32.0 / 33.5	7.6 / 8.3	19.8 / 20.9
✗	✗	✗	✓	33.2 / 34.5	7.0 / 7.6	20.1 / 21.1
✓	✗	✗	✗	34.8 / 36.2	14.1 / 15.1	24.5 / 25.7
✓	✗	✗	✓	35.1 / 36.4	13.0 / 14.1	24.1 / 25.3
✓	✗	✓	✓	35.1 / 36.3	14.4 / 15.4	24.8 / 25.9
✓	✓	✗	✗	35.4 / 36.6	15.2 / 16.1	25.3 / 26.4
✓	✓	✓	✓	35.1 / 36.3	16.1 / 17.1	25.6 / 26.7

Table 3. Ablation study of VETO on VG. **P**: Entity Part-level Patch Generation; **CR**: Cross-Relation Patch Generation; **CM**: Cross-Modality Patch Fusion; **D**: Depth.

Model	PredCls		
	R@k: 50/100	mR@k: 50/100	A@k: 50/100
VTransE	55.7 / 57.9	14.0 / 15.0	34.9 / 36.5
VTransE + GCL	35.5 / 37.4	30.4 / 32.3	33.0 / 34.9
VTransE + **MEET** (ours)	55.4 / 60.9	15.2 / 21.7	35.3 / 41.3
Motifs	65.3 / 66.8	16.4 / 17.1	40.9 / 42.0
Motifs + GCL	44.5 / 46.2	36.7 / 38.1	40.6 / 42.2
Motifs + **MEET** (ours)	63.6 / 68.4	25.0 / 30.4	44.3 / 49.4
VCTree	63.8 / 65.7	16.6 / 17.4	40.2 / 41.6
VCTree + GCL	44.8 / 46.6	35.4 / 36.7	40.1 / 41.7
VCTree + **MEET** (ours)	57.3 / 63.7	26.1 / 31.3	41.7 / 47.5
SHA	63.3 / 65.2	19.5 / 21.1	41.4 / 43.2
SHA + GCL	42.7 / 44.5	41.0 / 42.7	41.9 / 43.6
SHA + **MEET** (ours)	69.7 / 74.4	34.2 / 42.3	52.0 / 58.4
VETO (ours)	65.5 / 66.9	22.2 / 23.1	43.9 / 45.0
VETO + MEET (ours)	**73.9 / 78.3**	**43.3 / 50.5**	**58.6 / 64.4**

Table 2. Recall (R), mean Recall (mR), and their average (A) on GQA (the higher, the better). The best results before MEET debiasing are underlined and after MEET debiasing are highlighted in **bold.**



Plot 2. R@100 improvement on PredCls for VETO + MEET over SHA + MEET. The predicates are sorted based on their frequency in descending order.

References & Funding

[1] Zellers, Rowan, et al. "Neural motifs: Scene graph parsing with global context." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
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[4] Dong, Xingning, et al. "Stacked hybrid-attention and group collaborative learning for unbiased scene graph generation." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

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